

TollFreeForwarding Product Requirements Document (PRD)

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1.0.0

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1. Executive Summary & Vision

TollFreeForwarding is a next-generation, cloud-based virtual phone number management and smart PBX platform (SaaS). It empowers modern businesses to establish a global presence instantly by provisioning local and toll-free numbers, building complex interactive voice response (IVR) call flows, and monitoring inbound/outbound communications without relying on physical telephony hardware.

1.1 Core Value Proposition

- **Instant Provisioning:** Immediate setup of virtual phone numbers across global regions (US, UK, EU, APAC).
- **Smart Call Routing:** Advanced timezone-aware, scheduling-based, and caller ID routing.
- **No-Code IVR Builder:** Visual drag-and-drop canvas to configure nodes for voicemail, forwarding, menus, and call rejection.
- **Unified Messaging & Communications:** Support for inbound and outbound SMS/MMS, call logs, and call recordings.
- **Multi-tenant SaaS Architecture:** Built-in billing, subscriptions, role-based controls, and an admin workspace.

2. Platform Architecture

The system is organized as a unified monorepo deploying both backend and frontend applications onto Vercel.

tollfreeforwarding/	
% % % backend/	Node.js + Express + TypeScript
% % % % prisma/schema.prisma	Database schema (Neon PostgreSQL)
% % % % src/	
% % % % index.ts	Express app entry + CORS + proxy config
% % % % lib/	Auth, validate, cache, audit, API error, async handlers
% % % % routes/	Auth, numbers, routing, logs, dashboard, webhooks, SMS, Stripe
% % % % services/	Auth, numbers, billing, routing, email, SMS
% % % frontend/	React 18 + Vite + TypeScript + Tailwind CSS
% % % % src/	
% % % % pages/	Authentication, Dashboard, Numbers, Routing Rules, IVR Builder, SMS, Admin
Panel	
% % % % stores/	Auth and Theme stores (Zustand with persistence)
% % % % components/	AppLayout, MarketingLayout, UI library
% % % api/index.ts	Vercel Serverless Function entry shim

3. Database Schema

The persistence layer uses a Neon serverless PostgreSQL database managed via Prisma ORM. Below is the schema layout:

3.1 User (users table)

Stores user accounts, subscriptions, metadata, and plan limits.

- **id** (UUID, Primary Key): Unique identifier.
- **email** (String, Unique): User email address.
- **passwordHash** (String): Securely hashed bcrypt password.
- **firstName** / **lastName** (String): User's profile details.
- **plan** (String, Default: **STARTER**): Subscription level (**STARTER**, **PROFESSIONAL**, **ENTERPRISE**).
- **role** (String, Default: **user**): Access role (**user**, **admin**).
- **resetToken** / **resetTokenExpires** (String/DateTime, Nullable): Forgot password tokens.
- **stripeCustomerId** / **stripeSubscriptionId** (String, Nullable): Stripe integration keys.
- **trialEndsAt** (DateTime, Nullable): Evaluation period limit.
- **minutesUsedThisPeriod** (Integer, Default: 0): Monthly call quota tracking.
- **billingPeriodStart** (DateTime, Nullable): Billing cycle start date.

3.2 Virtual Number (**virtual_numbers** table)

Virtual numbers assigned to users, linked to either Mock or real Twilio providers.

- **id** (UUID, Primary Key): Unique identifier.
- **userId** (UUID, Foreign Key): Links to **User**.
- **e164Number** (String, Unique): E.164 formatted phone number.
- **friendlyName** (String, Nullable): Human-readable name.
- **countryCode** (String): ISO 2-letter country code.
- **numberType** (String, Default: **LOCAL**): Call category (**LOCAL**, **TOLL_FREE**).
- **status** (String, Default: **ACTIVE**): **ACTIVE** or **SUSPENDED**.
- **timezone** (String, Default: **America/New_York**): Working timezone.
- **voicemailGreeting** (String, Nullable): TTS or media greeting.
- **ivrEnabled** (Boolean, Default: **false**): Toggles IVR engine routing.
- **ivrFlow** (JSON String, Nullable): Node-flow configuration for the IVR engine.
- **providerSid** (String, Nullable): Twilio / provider SID.
- **provider** (String, Default: **mock**): Underlying telephony provider (**mock** or **twilio**).
- **deletedAt** (DateTime, Nullable): Timestamp for soft-delete recycling.

3.3 Routing Rule (**routing_rules** table)

Routing rules evaluated sequentially when IVR is disabled.

- **id** (UUID, Primary Key): Unique identifier.
- **virtualNumberId** (UUID, Foreign Key): Links to **VirtualNumber**.
- **priority** (Integer, Default: 0): Order of execution.
- **label** (String): Rule descriptor.
- **activeDays** (String, Default: **1,2,3,4,5**): Days active (1 = Monday, etc.).
- **openTime** / **closeTime** (String, Nullable): Time windows (HH:MM format).
- **action** (String): Routing endpoint category (**FORWARD_PSTN**, **FORWARD_SIP**, **RING_GROUP**, **VOICEMAIL**, **REJECT**).
- **destinations** (JSON Array, Default: **[]**): Targets for forwarding or ring groups.
- **ringStrategy** (String, Default: **SEQUENTIAL**): Call distribution method (**SEQUENTIAL** or **SIMULTANEOUS**).
- **ringTimeout** (Integer, Default: 30): Rings duration in seconds.
- **sipUri** (String, Nullable): Direct SIP target.
- **ivrNodeKey** (String, Nullable): Map-link node pointer.
- **deletedAt** (DateTime, Nullable): Timestamp for soft-delete recycling.

3.4 Call Log (**call_logs** table)

Session records of inbound and outbound calls.

- **id** (UUID, Primary Key): Unique identifier.

- **virtualNumberId** (UUID, Foreign Key): Links to **VirtualNumber**.
- **userId** (UUID, Foreign Key): Links to **User**.
- **providerCallSid** (String, Nullable): Call unique ID from Twilio.
- **direction** (String, Default: **INBOUND**): **INBOUND** or **OUTBOUND**.
- **callerNumber** / **calledNumber** (String): Source and target numbers.
- **forwardedTo** (String, Nullable): Final forwarding destination.
- **status** (String): Call resolution (**COMPLETED**, **BUSY**, **NO_ANSWER**, **FAILED**).
- **duration** (Integer, Nullable): Call duration in seconds.
- **recordingUrl** (String, Nullable): Live MP3 call recording location.
- **voicemailUrl** (String, Nullable): Voicemail message path.
- **transcription** (String, Nullable): Automated voicemail-to-text string.
- **startedAt** / **endedAt** (DateTime): Lifecycle timestamps.

3.5 SMS Log (**sms_logs** table)

Complete repository of SMS and MMS text conversations.

- **id** (UUID, Primary Key): Unique identifier.
 - **virtualNumberId** (UUID, Foreign Key): Links to **VirtualNumber**.
 - **userId** (UUID, Foreign Key): Links to **User**.
 - **direction** (String): **INBOUND** or **OUTBOUND**.
 - **fromNumber** / **toNumber** (String): Active numbers.
 - **body** (String): Text message payload.
 - **status** (String): Delivery status (**RECEIVED**, **SENT**, **FAILED**).
 - **providerSid** (String, Nullable): Telephony provider SID.
 - **mediaUrls** (String, Nullable): Array of image/attachment URLs.
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4. Key Product Features

4.1 Secure Authentication & Session Management

- **JWT Access & Refresh Rotation:** Uses a secure 15-minute JWT Access token and a rotating 7-day Refresh token stored in cookies to minimize exposure risks.
- **API Key Infrastructure:** Developers can generate prefix-masked API Keys securely hashed using SHA-256 in the database to integrate third-party webhooks.
- **Security Controls:** Absolute protection against timing-based login enumeration by standardizing bcrypt execution paths on non-existent accounts.

4.2 Virtual Number Provisioning & Lifecycle

- **Twilio Integration:** Connects directly with real-time Twilio local/toll-free endpoints.
- **Soft-Delete Recycle Bin:** Deleted numbers remain restorable within a 30-day window, preventing accidental loss of client-facing lines.
- **Limits Enforcement:** Plans enforce hard gates on provisioned lines: STARTER allows 3 numbers, PROFESSIONAL allows 5 numbers, and ENTERPRISE permits unlimited lines.

4.3 Advanced Telephony Routing

- **Timezone & Time-of-Day Gates:** Utilizes Luxon to translate rule configurations relative to the timezone of the virtual number, routing after-hours calls to voicemail or backup lines.
- **Ring Groups & Strategy:** Distributes calls sequentially or simultaneously to multiple phone numbers using custom timeouts.
- **TwIML Generation Engine:** Dynamic generator returns compliant XML payloads (**<Dial>**, **<Sip>**, **<Record>**, **<Hangup>**) for inbound webhooks.

4.4 Visual Drag-and-Drop IVR Builder

- **React Flow Interface:** Users construct call flows using drag-and-drop nodes on a workspace canvas.

- **Interactive IVR Engine:** Executes node traversals on incoming webhook triggers:
 - **Greeting Node:** Text-to-speech output.
 - **Menu Node:** Plays greetings and monitors DTMF digit selections.
 - **Forward Node:** Dynamically transfers the call.
 - **Voicemail Node:** Records the caller and uploads to secure storage.
 - **Hangup Node:** Terminates the call.

4.5 Subscription Billing & Usage Metering

- **Stripe Billing Portal:** Managed billing flow (Stripe Checkout & Billing Portal) for subscriptions.
 - **Minutes Metering:** Automatically tracks consumed minutes and sends notification alerts when accounts near their monthly limits.
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5. System API Routes

5.1 Authentication API

- **POST /api/auth/register** - Creates a new user profile.
- **POST /api/auth/login** - Authenticates user and sets session cookies.
- **POST /api/auth/refresh** - Rotates refresh tokens and returns fresh access tokens.
- **POST /api/auth/logout** - Revokes refresh tokens and clears session data.
- **POST /api/auth/forgot-password** - Dispatches password reset tokens via Resend.
- **POST /api/auth/reset-password** - Resets profile passwords using a valid reset token.
- **GET /api/auth/me** - Retrieves session user details.
- **PATCH /api/auth/profile** - Modifies profile details.
- **PATCH /api/auth/password** - Updates account passwords.

5.2 API Key Management

- **GET /api/auth/api-keys** - Lists active API Keys (masked).
- **POST /api/auth/api-keys** - Generates a new API Key prefix.
- **DELETE /api/auth/api-keys/:id** - Revokes an API Key.

5.3 Virtual Numbers API

- **GET /api/numbers** - Retrieves a paginated list of active numbers.
- **POST /api/numbers** - Provisions local or toll-free numbers.
- **GET /api/numbers/:id** - Returns metadata and routing rules.
- **PATCH /api/numbers/:id** - Modifies name, timezone, or toggles IVR settings.
- **DELETE /api/numbers/:id** - Moves a virtual number to the soft-delete recycle bin.
- **GET /api/numbers/recycle-bin** - Lists numbers currently in the recycle bin.
- **POST /api/numbers/:id/restore** - Restores a soft-deleted virtual number.

5.4 Call Logs & Analytics

- **GET /api/logs** - Retrieves filtered and paginated call history.
- **GET /api/dashboard** - Computes call statistics, active call rates, and trend data.

5.5 SMS Suite API

- **GET /api/sms/threads** - Retrieves structured conversation logs sorted by latest interaction.
- **GET /api/sms/threads/:phoneNumber** - Lists message history for a specific sender.
- **POST /api/sms/send** - Dispatches outbound messages via the Twilio integration.

5.6 Admin Operations API

- **GET /api/admin/users** - Lists all platform users.
- **PATCH /api/admin/users/:id** - Modifies a user's system plan, active role, or billing state.
- **POST /api/admin/users/:id/reset-minutes** - Resets active usage counters.
- **GET /api/admin/audit-logs** - Inspects system audit logs.

5.7 Telephony & Webhooks

- **POST /webhook/inbound** - Main inbound Twilio handler verifying signatures and returning TwiML configurations.
- **POST /webhook/stripe** - Decodes Stripe checkout session completions and subscription changes.